Calvin M. Bailey

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Education

Denison University | Granville, OH — Graduated: 05/2025

Paris Lodron Universität Salzburg | Salzburg, AT - Semester Abroad — 03/2024 - 07/2024

***Bachelor of Arts**:* Data Analytics (Environmental Studies Concentration) | ***Bachelor of Arts**:* German

- **Relevant coursework:** Data for Political Research | Data Systems | Applied Statistics | Advanced Predictive Analysis | Practicum in Data Analytics | Data Analytics Senior Seminar | Environmental Analysis | Applied GIS for Earth and Environmental Science
- Dean's List Fall 2023, (CEFR) Seal of Biliteracy in German

Skills

- **Coding:** ArcGIS Pro, QGIS, R, Python **Languages:** English (Native), German (B1)

Work Experience

Bucknell University |Lewisburg, PA — 06/2025 - 10/2025

Forestry Data Science Research Fellow

- Conducted simulation-based evaluation of 10 different small area estimators to assess performance in recently burned forest regions Across 4 states with sparse sampling, informing methodology choices for the U.S. Forest Service Forest Inventory and Analysis Program.
- Part of a National effort to support the Forest Service in meeting a Congressional directive to evolve the nation's small area estimation capacity for forest inventory.
- Provided a presentation for stakeholders focusing on the five most promising performers, a Quarto document explaining all code within R, and a Research Paper.

Maine Sport Outfitters|Rockport, ME / Hybrid — 11/2025 - 11/2026

Database Manager

- Upheld organization within Lightspeed P.O.S system to ensure smooth dealer to customer online and in-store experience.
- Ensured the efficiency and cleanliness of new product creation. While reorganizing past systems to decrease errors for the current team.

Projects

Data Analytics 301 Final Project|Denison University — Fall 2024

Forecasting Sales for PANG Industrial: Correlation, Time Series, and External Factor Modeling

- Analyzed global and U.S. sales data using Python and R to forecast trends with SARIMA/SARIMAX models; identified key economic predictors (e.g., ACMSNO, unemployment rate, oil prices).
- Integrated external data (FRED, EM-DAT, MIT Election Lab) to assess impacts of disasters and policy; developed actionable insights for strategic forecasting and dashboard integration.

Data Analytics 401 Capstone Project| Denison University — Spring 2025

Geospatial Growth Viability Analysis of Pawpaw in Rural Food Insecure Areas

- Designed and implemented GIS-based suitability models in ArcGIS Pro to evaluate pawpaw cultivation in USDA-defined rural food deserts; integrated raster data (soil, slope, irradiance, precipitation, land use) with food access indicators.
- Identified high-scoring orchard plots in Virginia and Oklahoma counties; estimated over 110,000 lbs of annual yield, proposing a replicable framework for native, low-input fruit interventions in underserved regions.

On-Campus Leadership and Involvement

Denison Climbing Club — 09/2023 - 05/2025

Co-Founder and Treasurer

- Exhibited organizational skills by creating a budget and spending objectives to increase membership, participation, and club culture.
- Collaborated with Denison Athletic Department and Denison Campus Government Association to secure necessary funding and resources while overseeing all potential spending and spending goals.